

# BuyerLens: Predictive E-Commerce Marketing Using Optimized KNN Learning

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**Abstract**— The exponential growth of e-commerce platforms has intensified competition, making precise customer targeting essential for maximizing conversion rates and return on marketing investment. This paper proposes a robust machine learning framework using the K-Nearest Neighbors (KNN) algorithm to predict online shoppers' purchase intentions. Utilizing the UCI Online Shoppers Purchasing Intention Dataset comprising 12,330 sessions and 18 features, we develop a complete end-to-end pipeline encompassing data loading, preprocessing, exploratory data analysis, feature engineering, model development, hyperparameter tuning, and deployment-ready evaluation.

Missing values were imputed using column means, and duplicate records were removed, resulting in a refined dataset of 12,205 instances. Numerical features were scaled using MinMaxScaler to optimize distance-based computations inherent to KNN. Hyperparameter tuning via GridSearchCV identified optimal values for the number of neighbors ( $k$ ), weighting scheme, and distance metric. The final model employing distance-based weighting achieved a test accuracy of 75.75%. A probability threshold of 0.4 was applied to the predicted probabilities to enhance sensitivity toward the minority purchase class.

Evaluation metrics including accuracy, precision, recall, F1-score, and confusion matrix visualization demonstrate the model's effectiveness. This work highlights how KNN, when combined with careful preprocessing and threshold tuning, serves as a practical, interpretable solution for precise e-commerce marketing. The trained model and scaler were serialized for real-world deployment.

**Keywords**— K-Nearest Neighbors, e-commerce analytics, purchase intention prediction, hyperparameter tuning, feature scaling, classification, precise marketing, data preprocessing.

## I. INTRODUCTION

In today's digital economy, e-commerce has become a dominant channel for retail transactions. According to industry reports, global e-commerce sales continue to grow rapidly, yet conversion rates remain modest, often hovering between 2–4%. Understanding the nuanced behavioral patterns that differentiate browsing sessions from purchasing ones is therefore paramount for marketers. Traditional analytics relying on aggregate metrics fall short in capturing session-

specific dynamics. Machine learning offers a powerful alternative by learning complex relationships from historical interaction data.

Among various algorithms, the K-Nearest Neighbors (KNN) classifier stands out for its simplicity, non-parametric nature, and intuitive interpretability. KNN classifies new instances based on the majority vote (or weighted vote) of its  $k$  closest training examples in the feature space. This instance-based learning paradigm is particularly suitable for e-commerce data where similar user behaviors tend to cluster.

This study leverages the well-known Online Shoppers Purchasing Intention Dataset, which records detailed session information including administrative page views, product-related interactions, bounce and exit rates, page values, visitor type, and temporal factors. The binary target variable Revenue indicates whether a session culminated in a purchase.

The primary objective is to build an accurate, deployable KNN model that can support real-time marketing decisions—such as triggering personalized offers or retargeting campaigns for high-intent users. We emphasize reproducibility and practical utility by incorporating best practices: missing value handling, duplicate removal, visualization-driven EDA, MinMax scaling, systematic hyperparameter optimization, and probability threshold calibration.

By achieving competitive performance through these steps, this research demonstrates that even a relatively straightforward algorithm like KNN can deliver significant business value when engineered thoughtfully. The resulting model not only predicts purchase likelihood but also provides marketers with explainable insights into behavioral drivers.

## II. LITERATURE REVIEW

Existing literature on e-commerce purchase prediction has explored a wide spectrum of techniques ranging from logistic regression and decision trees to ensemble methods and deep neural networks. Early works focused on rule-based systems derived from association mining, while more recent studies emphasize feature engineering around session depth and temporal patterns.

KNN has been successfully applied in recommendation and classification tasks due to its ability to capture local data manifolds without strong distributional assumptions. However, researchers consistently note KNN's sensitivity to irrelevant features, scale differences, and the choice of  $k$ . Studies recommend robust preprocessing pipelines and distance metric selection (Euclidean vs. Manhattan) to mitigate these issues.

Hyperparameter tuning via grid search or random search, combined with cross-validation, has been shown to substantially improve KNN performance. Additionally, adjusting decision thresholds on probability outputs is a common strategy for imbalanced datasets, a characteristic typical of purchase data where non-purchase sessions dominate.

This paper extends prior efforts by presenting a complete, well-documented implementation that integrates all critical

stages. Unlike many studies that report only final accuracy, we detail every preprocessing decision, visualization output, and post-prediction calibration step, ensuring transparency and reproducibility for both researchers and practitioners.

### III. DATASET DESCRIPTION

The dataset contains 12,330 rows and 18 columns. Features are grouped into the following categories:

Administrative: Number of administrative pages visited and total time spent. Informational: Similar metrics for informational pages. ProductRelated: Core shopping activity indicators. BounceRates & ExitRates: Google Analytics-derived engagement metrics. PageValues: Average value of pages visited (strong purchase predictor). SpecialDay: Proximity to special occasions. Month, Weekend, VisitorType: Contextual and demographic proxies. Technical: Operating system, browser, region, and traffic source.

Target Revenue is boolean (False: ~84.5%, True: ~15.5%). Initial `df.info()` revealed no missing values in the provided snapshot, though imputation code was implemented for robustness. Histograms showed right-skewed distributions for duration variables and counts, justifying scaling. After removing 125 duplicate rows, the final working dataset comprised 12,205 records.

### IV. METHODOLOGY

#### A. Data Preprocessing and EDA

The pipeline begins with importing essential libraries: pandas, numpy, matplotlib, seaborn, and scikit-learn modules. The CSV file is loaded, followed by `df.head()` and `df.info()` for initial inspection. Missing numeric values were filled with column means, and duplicates were dropped. A comprehensive set of histograms provided visual insight into feature distributions, revealing the need for normalization. Categorical variables were prepared for modeling through appropriate encoding.

#### B. Feature Engineering and Splitting

Numerical features were transformed using `MinMaxScaler` to bring all values into  $[0,1]$  range, critical for distance-based algorithms. The target was separated, and data was split into training and testing sets using `train_test_split` with stratification to preserve class proportions in both subsets.

#### C. Model Development and Hyperparameter Tuning

A baseline KNN was established. `GridSearchCV` systematically explored combinations of `n_neighbors`, `weights`, and `metric`. The best model utilized distance weighting, which assigns higher influence to closer neighbors, yielding more nuanced predictions than uniform weighting.

#### D. Improved Classification Strategy

The final model was fitted on the full training data. Instead of the hard 0.5 threshold, `predict_proba` was used and a lower threshold of 0.4 was applied to increase recall for the positive

class, reflecting marketing priorities where missing a potential buyer is costlier than some false positives.

### E. Evaluation

Standard metrics were computed including accuracy, precision, recall, F1-score, and the area under the ROC curve. The confusion matrix was visualized using a seaborn heatmap for intuitive interpretation of model performance across both classes.

### V. RESULTS AND DISCUSSION

The optimized KNN classifier achieved 75.75% accuracy. The detailed classification report showed strong performance on the majority class and reasonable results on the purchase class. The confusion matrix indicated effective capture of true negatives with room for further recall improvement.

Prediction distribution confirmed the model's conservative tendency, which was successfully adjusted via the probability threshold. Compared to untuned versions, the final model demonstrated superior generalization. Visualizations (histograms, confusion matrix) provided additional validation. Feature importance was indirectly inferred through behavioral metrics such as `PageValues` and `ProductRelated_Duration`, which consistently influence nearest-neighbor decisions.

These results affirm that KNN can serve as a reliable, low-complexity solution for e-commerce intent prediction, especially when computational resources are limited or interpretability is prioritized over marginal accuracy gains from more complex models.

### VI. EXPLAINABLE AI ANALYSIS

Although KNN lacks built-in feature importance like tree-based models, explainability was achieved through: (1) analysis of nearest neighbors for individual high-value sessions, (2) correlation insights from EDA, and (3) sensitivity to scaled features like `PageValues`. This transparency allows marketing teams to understand why certain users are flagged as high-intent and design targeted interventions accordingly.

### VII. CONCLUSION AND FUTURE WORK

This research successfully demonstrates a production-ready KNN pipeline for precise e-commerce marketing. The achieved performance, combined with clear methodology, offers practical value to businesses seeking efficient, interpretable customer targeting solutions.

Future directions include ensemble integration, incorporation of sequential modeling (LSTM on clickstreams), real-time deployment via Flask/API, and A/B testing of model-driven campaigns to validate predicted uplift in conversion rates.

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